

Unit

**Algorithms and Mathematical
Concepts for computing**

Title

**Optimizing Supply Chain Logistics
Using Algorithm Development and
Mathematical Concepts**

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Name: Min Hein Kyaw

Student ID: 218542

Name of the Centre: Twinkle University

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Preface

This report will describe how to optimize logistic system by the various algorithm.

It shows how to use algorithms, predictive models, visual representation tools and how to manage the data sets.



Figure 1. Supply chain logistic (FREEP!K, 2025)

Task 1

a) Meaning of proposition

It is a statement that shows whether it is TRUE or FALSE. It can be used in the system to express the situations of various conditions.

Proposition A: The warehouse has enough stock.

Proposition B: ERP system is online and provide real-time data.

Proposition C: The weather is fine.

Proposition D: The road is good condition for transportation.

Proposition E: Delivery address is correct and confirmed.

Proposition F: The deadline of delivery is much enough for transport.

Proposition G: Vehicle is available to transport.

Proposition H: There are alternative ways to transport. (e.g., air, sea, rail).

(MECALUX, 2025)

b) Pseudocode formulation

Pseudocode is used for choosing the best routes by using the propositions that are mentioned above.

Start

IF A = TRUE AND B = TRUE THEN

 IF C = TRUE AND D = TRUE THEN

 IF E = TRUE AND F = TRUE THEN

 IF G = TRUE AND H = TRUE THEN

 select_optimal_route();

 ELSE

 OUTPUT "Route selection failed ";

 END IF;

 ELSE

 OUTPUT "Route selection failed ";

 END IF;

 ELSE

 OUTPUT "Route selection failed ";

 END IF;

ELSE

 OUTPUT "Route selection failed ";

END IF

END

c) Constraints and Another way to solve

i. Limitations In route decision (Propositional logic)

a. Only have TRUE and FALSE value

It only shows the TRUE and FALSE condition of the statement.

Example: IF vehicle is not available, it cannot describe why it is not available.

b. Do not represent proper amount

It cannot show the quantity of things such as all or some.

Example: Although the warehouse has enough stock, it doesn't describe the exact amount of stock in quantity.

c. Limited language process

It does not understand the language that the user put in.

Example: When ERP system is wanted to know, it does not explain about what the ERP system is.

d. Lack of expression of time

It cannot describe the exact time of the deadline.

Example: If the deadline is near, it only shows it is not possible instead of describe the date that it can be delivered.

e. No relationship

It cannot express the relationship of the statements.

Example: When the road is good condition and vehicle is ready, it cannot describe that vehicle can drive due to the safe road. (geeksforgeeks., 2025)

ii. Other ways to solve problems

Heuristic algorithms

It gives solution of any complex statement in the easier and faster manner. (Srinidhi, 2019)

Constraint satisfaction techniques

It contains stating problems and finding solutions that are satisfied for the statements. (Neagu, 2004)

By using the above two techniques,

a. Do not show only TRUE and FALSE value

It can show the reasons of the statement.

Example: IF vehicle is not available, it can show why it is not available.

b. Represent proper amount

It can show the quantity of things such as all or some.

Example: If the warehouse has enough stock, it can describe the exact amount of stock in quantity.

c. No limited language process

It can understand the language that the user put in.

Example: When user wants to know about the ERP system, it can explain about what the ERP system is.

d. Expression of time

It can describe the exact time of deadline.

Example: If deadline is near, it can describe the date that it can be delivered.

e. Express relationship

It can express the relationship of the statements.

Example: When the road is safe and vehicle is ready, it can describe that vehicle can drive due to safe road.

d) Overview and summary

In summary, pseudocode can be applied to find the optimal routes by using propositions but there are many limitations as it can only describe TRUE or FALSE value. So, it is not efficient in real life problems. For solving the limitations, Heuristic algorithms or constraint satisfaction techniques are available to use. They can express not only true and false but also the reasons for the users. So, it is very efficient and nowadays they are used together in various situations.

In supply chain logistics, selecting the optimal routes is important because the goods need to delivery efficiently, and in a timely manner. So, it can enhance customers satisfaction and improve the system.

Task 2

a) Predicate

Predicate is the extension of propositional logic that contains variables which can be substituted. They express the relationships and properties among objects. (geeks, 2025)

Predicates can be used in the system and impact on demand forecasting.

S(k): Season based changes affect demand for item k.

P(k): Processing period impact demand for item k.

E(k): Economic situations influence demand for item k.

M(k): Market Trends impact demand for item k.

T(k): Tech progress affects demand for item k.

A(k): Advertising efforts impact demand for item k.

H(k): Historical Sales Data affect demand for item k.

U(k): Uncontrollable factors (such as natural disaster) impact demand for item k.

C(k): Competitive moves influence demand for item k.

CC(k): Customers confidence influences demand for item k. (WorkTrek, 2025)

b) Quantifiers

They can be used to express the property of elements in the domain. They can be applied with predicates that show the overall demand trends as below.

$$\forall x \text{ (item } k): S(k) \wedge M(k) \rightarrow \text{forecast_increase}(k)$$

$$\exists x \text{ (item } k): S(k) \wedge A(k) \rightarrow \text{forecast_increase}(k)$$

$$\forall x \text{ (item } k): P(k) \wedge C(k) \rightarrow \text{forecast_increase}(k)$$

$$\exists x \text{ (item } k): E(k) \wedge M(k) \rightarrow \text{forecast_increase}(k)$$

$$\forall x \text{ (item } k): E(k) \wedge CC(k) \rightarrow \text{forecast_increase}(k)$$

$$\exists x \text{ (item } k): T(k) \wedge C(k) \rightarrow \text{forecast_increase}(k)$$

$$\forall x \text{ (item } k): H(k) \wedge A(k) \rightarrow \text{forecast_increase}(k)$$

$$\exists x \text{ (item } k): CC(k) \wedge P(k) \rightarrow \text{forecast_increase}(k)$$

$$\forall x \text{ (item } k): T(k) \wedge \neg U(k) \rightarrow \text{forecast_increase}(k)$$

$$\exists x \text{ (item } k): U(k) \wedge \neg H(k) \rightarrow \text{forecast_decrease}(k)$$

c) Explanation of Predictive Modelling

It is used to create mathematical model that predict future situation by previous data.
(geeks, 2025)

$$S(k) \wedge E(k) \rightarrow \text{forecast_stable}(k)$$

If there is seasonal trend for items but the ecological condition such as GDP is decline, then the demand of product k will be stable.

$$A(k) \wedge CC(k) \rightarrow \text{forecast_increase}(k)$$

If promotions present and customers' confidence (financial situation) is good, then the demand of item k will be increase.

$$U(k) \wedge \neg C(k) \rightarrow \text{forecast_decrease}(k)$$

If uncontrollable factor such as natural disaster occurred and competitive moves do *not* influence demand of item k, then demand of item k will be decrease.

$$P(k) \wedge H(k) \rightarrow \text{forecast_stable}(k)$$

If the duration of processing is short but there are no past sailing history records, then demand of item k will be stable.

$$M(k) \vee T(k) \rightarrow \text{forecast_increase}(k)$$

If there are shopping trends for the products and ERP system is well functioned, then demand of item k will be increase.

d) Overview and summary

In summary, predicate logic is propositional logic extension which used to express the relation and property of the objects. The quantifiers such as ($\forall$) and ($\exists$) can be used to express the property of elements in the domain. To predict the future outcome for the system, predictive modellings can be developed.

In supply chain logistic, forecasting demand of item can optimize the inventory levels. When the average demand of item is known, the inventory can calculate the reorder point and can restore item in time. So, item is always available. Knowing the demand the type of item, priority can be also calculated and storage cost can be reduced by not storing the unnecessary item.

Production planning can be improved by calculating based on the past data and future demand. Thus, the service level and quality of item will be improved. Item can be safely delivered in time and customer service level is enhanced. These are the facts that forecasting demand of item optimizes system.

Task 3

a) Usage of Venn Diagrams

It shows visual communication between sets.

S = Suppliers, M = Manufacturers, W = Warehouses,

D = Distribution centers,

A = Retail Shop A,

B = Retail Shop B

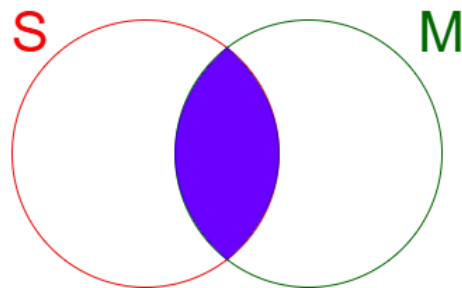


Figure 2. $S \cap M$

$S \cap M$ = Procurement from supplier effect manufacturing time

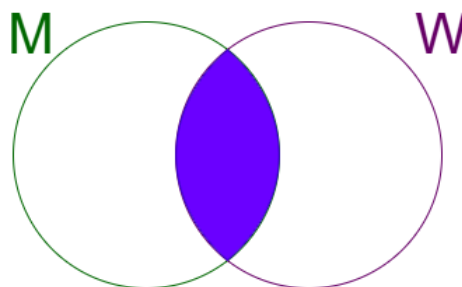


Figure 3. $M \cap W$

$M \cap W$ = Transportation from manufacturers maintains warehousing capabilities

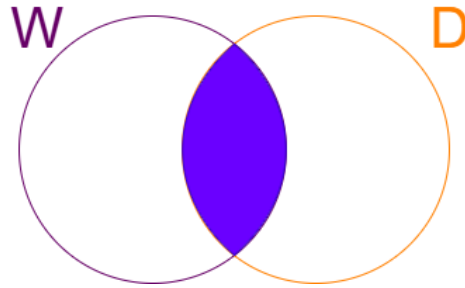


Figure 4. $W \cap D$

$W \cap D$ = inventory-management of warehouses supports enough stocks in distribution centers.

Alternative transportation ways

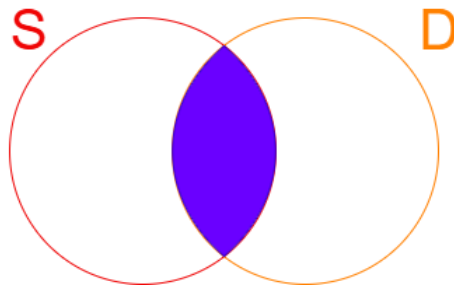


Figure 5. $S \cap D$

$S \cap D$ = 50 units are delivered weekly and need 5 days to received.

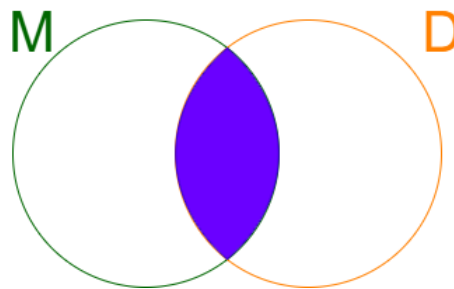


Figure 6. $M \cap D$

$M \cap D$ = 100 units are delivered weekly and need 10 days to received.

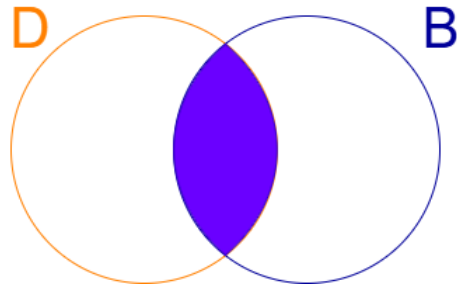


Figure 7. $D \cap B$

$D \cap B = 80$ units are delivered weekly and need 14 days to received.

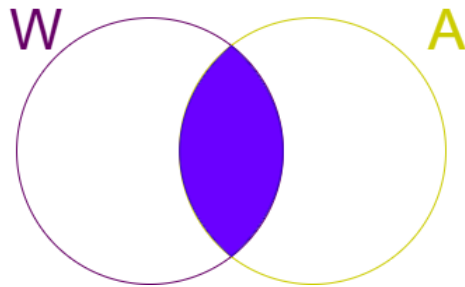


Figure 8. $W \cap A$

$W \cap A = 150$ units are delivered weekly and need 7 days to received.

b) Usage of various data structures

Graph

It shows the transportation route and can use to find the best routes to any location.

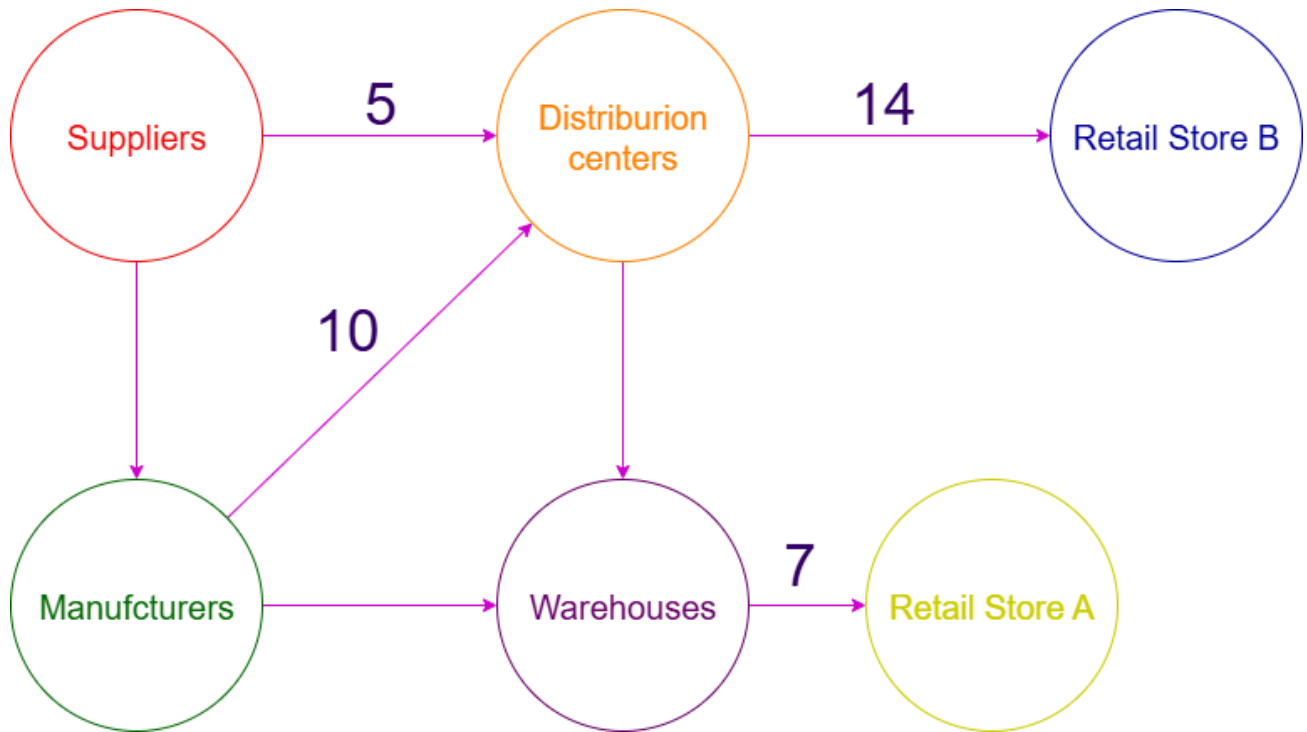


Figure 9. Graph showing transportation route with transit time (days).

Hash table

Products are stored in hash table as below.

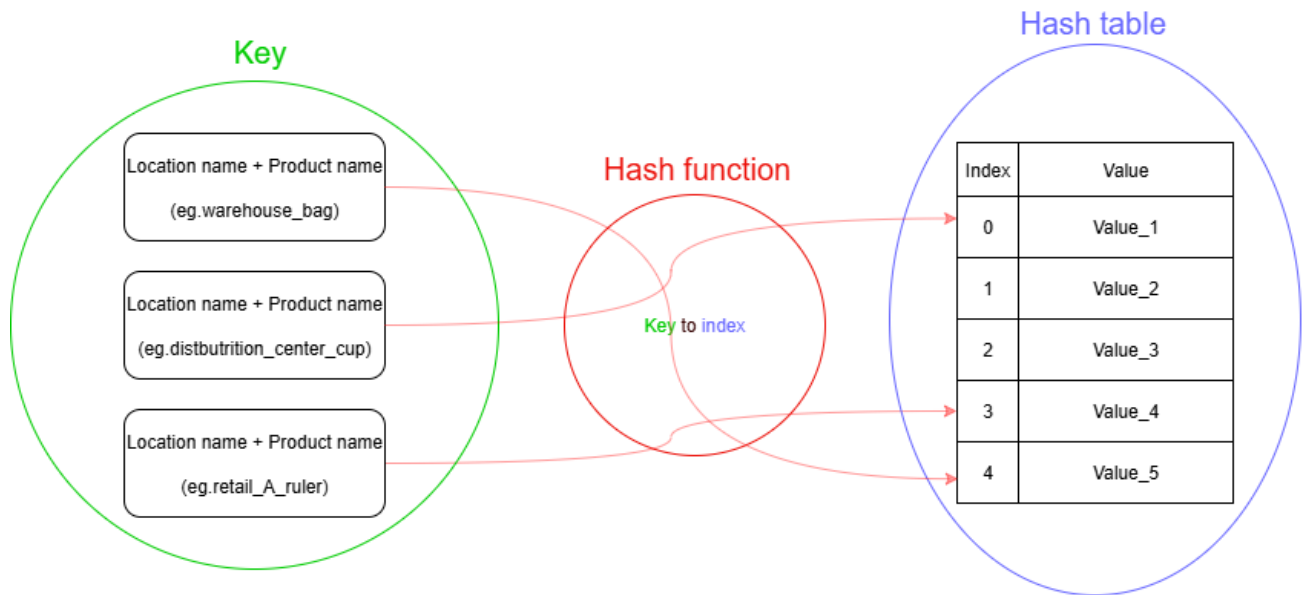


Figure 10. Storage management by hash table

Each value includes data such as quantity, quality and time.

Binary search tree (BST)

By using BST, **name**, **price**, **quantity** and **deadline** of products can be sorted and easily managed in each location.

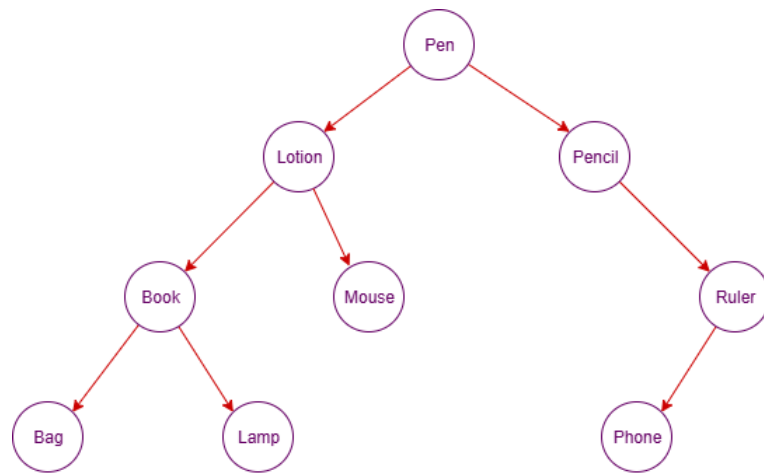


Figure 11. Sorting by name

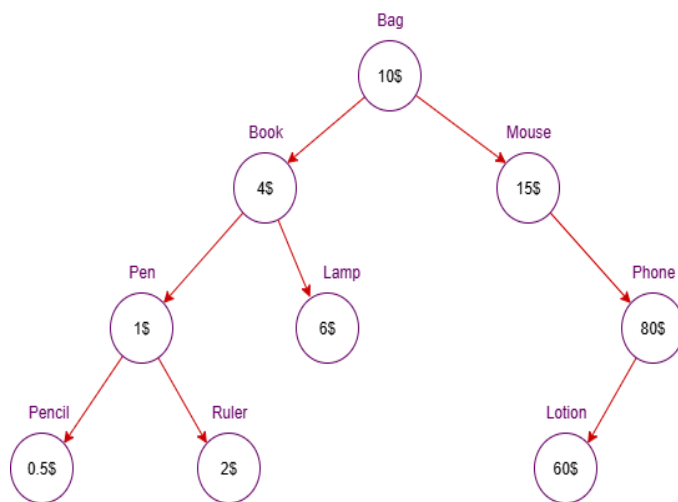


Figure 12. Sorting by price

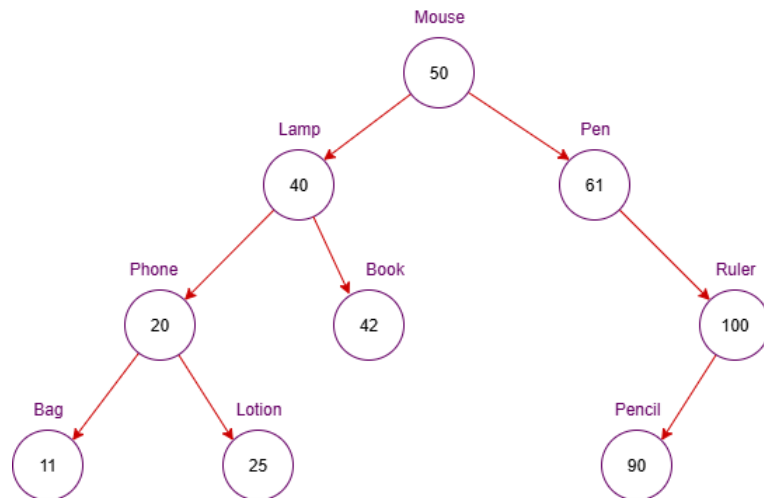


Figure 13. Sorting by quantity

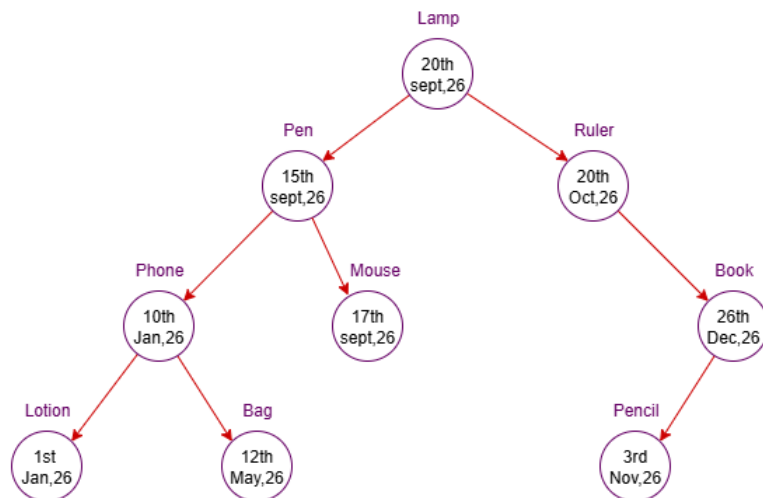


Figure 14. Sorting by deadline

c) Algorithms

Data structures are used by the algorithms as mentioned below.

Shortest path (Graph)

Step-1. Start

Step-2. Define current_location = Node S

Step-3. Define whole_distance = 0,

Step-4. Visit other nodes through direction.

- Note distances to other nodes from current node
- Compare them and take shorter distance
- Set new_distance = shorter distance
- Add new_distance to whole_distance
 $\text{whole_distance} = \text{whole_distance} + \text{new_distance}$
- Current_location = visit Node

Step-5. Repeat step 4 until there is no other node to visit.

Step-6. Output total_distance and this is the shortest part.

Step-7. End

Storage management (Hash table)

Step-1. Start

Step-2. Define inventorystore (location, product)

- Key = location + product

- Return Key

Step-3. Define Hash_table [0 to N]

Step-4. Define Hash_function (key).

- Index = 0

- for each word in key

 - Index = Index + ASCII

- Index = Index MOD N

- Return Index

Step-5. Input location and product name.

Step-6. Key = inventorystore (location, product)

Step-7. Index = Hash_function (key)

Step-8. Input value in Hash_table (Index)

Step-9. End

Sorting by quantity (BST)

Step-1. Start

Step-2. Define empty tree (root = Null)

Step-3. Input first product into root.

Step-4. Input another product.

 If value is smaller, go left node.

 If value is greater, go right node.

 Repeat until found empty node, then insert.

Step-5. Output product lists in order traversal.

Step-6. End

d) Overview and summary

In summary, Venn diagrams show visual relationship between locations. Graph can provide shortest path. Any product in any location can be searched easily and quickly by using hash table. (time complexity = $O(1)$). Products in each location can be searched easily without extra sorting by using binary search tree. (time complexity = $O(\log n)$).

Knowing shortest route, transportation time is being saved and cost is reduced. Products below reorder points can be easily found by using BST. Both overstocking and understocking can be detected easily and it is good for availability. ERP system can be used to integrate with above data to improve supply chain logistics.

Task 4

a) Searching & Sorting Algorithms

Quick sort

It is based on divide and conquer that chooses one unit as pivot and splits array around pivot. Then, it places pivot in its proper position in sorted array. It applies same technique to two partitioned sub-arrays and it only stops when single unit is remaining in sub array. (geeksforgeeks, 2025)

Merge sort

It follows divide and conquers. It splits array recursively into two until it can no more be split. Each subsection of an array is sorted individually. Then, all sorted subsections of an array are combined back together in sorted order. (geeksforgeeks, 2025)

Comparisons	Quick sort	Merge sort
Stability	Unstable	Stable
Sorting technique	Internal	External
Efficiency	Depend on pivot	Equally efficient for all types of inputs of the same size
Speed	Quick for small data set	Same speed on any size of data set

In inventory management, sorting algorithms are used in storing products to know about stock levels and search products easily.

Merge sort can be applied in sorting warehouse stock levels as there is a huge data set and the data set wants to be stable.

Quick sort can be used in sorting orders by delivery times as the data is small and needs to be faster. But the optimal pivot must be chosen.

Linear search

It searches each item in array until targeted item is identified or array is ended.
(geeksforgeeks, 2025)

Binary search

It repeatedly splits search interval in half until target item is identified or total search space is exhausted. (geeksforgeeks, 2025)

Comparisons	Linear search	Binary search
Must be sorted	No	Yes
Perform	Equality comparisons	Ordering comparisons
Efficiency	In small data sets	More complex
Speed	Very slow process	Very fast process
Used in	Multidimensional arrays	Only single dimensional array

In distribution, searching algorithms can be used to search targeted products.

Binary search can be used in searching products with ID numbers but the lists need to be sorted. It is a very fast process.

Linear search can be used in searching products in unsorted lists. Although it is very slow process, it can be used in multidimensional arrays.

b) Efficiency analysis

Sorting algorithms

Efficiency		Quick sort	Merge sort
Time complexity	Best case	$O(n \log n)$	$O(n \log n)$
	Average	$O(n \log n)$	$O(n \log n)$
	Worst case	$O(n^2)$ when pivot is chosen poorly	$O(n \log n)$
Space complexity		$O(n \log n)$	$O(n)$
Solution quality		Unstable, speed depend on pivot, better for small data set	Stable, faster, any type of data set

(code360, 2025)

In warehouse, merge sort can be used although more space consumption because it is stable.

In shipping with small data, quick sort can be used although it is unstable because it is quick.

Searching algorithms

Efficiency		Linear search	Binary search
Time complexity	Best case	$O(1)$	$O(1)$
	Average	$O(n)$	$O(\log n)$
	Worst case	$O(n)$	$O(\log n)$
Space complexity		$O(1)$	$O(1)$
Solution quality		Slow process, suitable for small data sets	Fast process, suitable for large data sets

(code360, 2025)

In distribution, binary search can be used in sorted lists and linear search can be use in unsorted lists because there is equal space consumption in both data.

c) Practical constraints

Computational resources

Central processing unit (CPU) – for calculations and decision making

Memory (RAM) – for storing data temporally, speed

Storage (HDD,SSD) – for store information and instructions

Graphics processing unit (GPU) – for image, animations and videos

For low CPU s, low GPU, and RAM, low space complexity algorithms such as quick sort need to be used.

Computers need to be high CPU, high GPU, high RAM and high data storage. So, ERP system can work well.

Scalability

Vertical scaling – Attach more source such as RAM, GPU, SSD to one computer

Horizontal scaling – Attach more computers to the system.

For scalability, low time complexity algorithms such as merge sort need to be used because it need to handle millions of records.

Cloud system can be used as it is easier and less cost than adding more computers.

Real time processing

Inventory management(ERP system) – to get update stock level.

Barcode scanner – to get item information.

GPS system – to get exact location.

The system needs low time complexity algorithms such as binary search because it needs fast response.

Internet speed also needs to be work well in this situation.

d) Overview and summary

In summary, the use of sorting and searching algorithms is based on time complexity, space complexity and also on the practical constraints such as computational resources.

For supply chain logistics, real time processing with large data sets are necessary. So, merge sorting algorithm and binary search algorithms are most suitable. In this case, practical constraints should be solved first.

Then, there will be efficient in inventory management, reduce delay in processing and the company can expand well. Finally, supply chain logistic is optimized and customer satisfaction is also improved.

Conclusion

By the above concepts, there will be lots of improvements such as choosing the optimal routes, timely delivery, cost reduction and efficient storage management. Finally, customer satisfaction is improved.

-----End of the Assignment-----

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